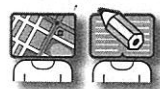
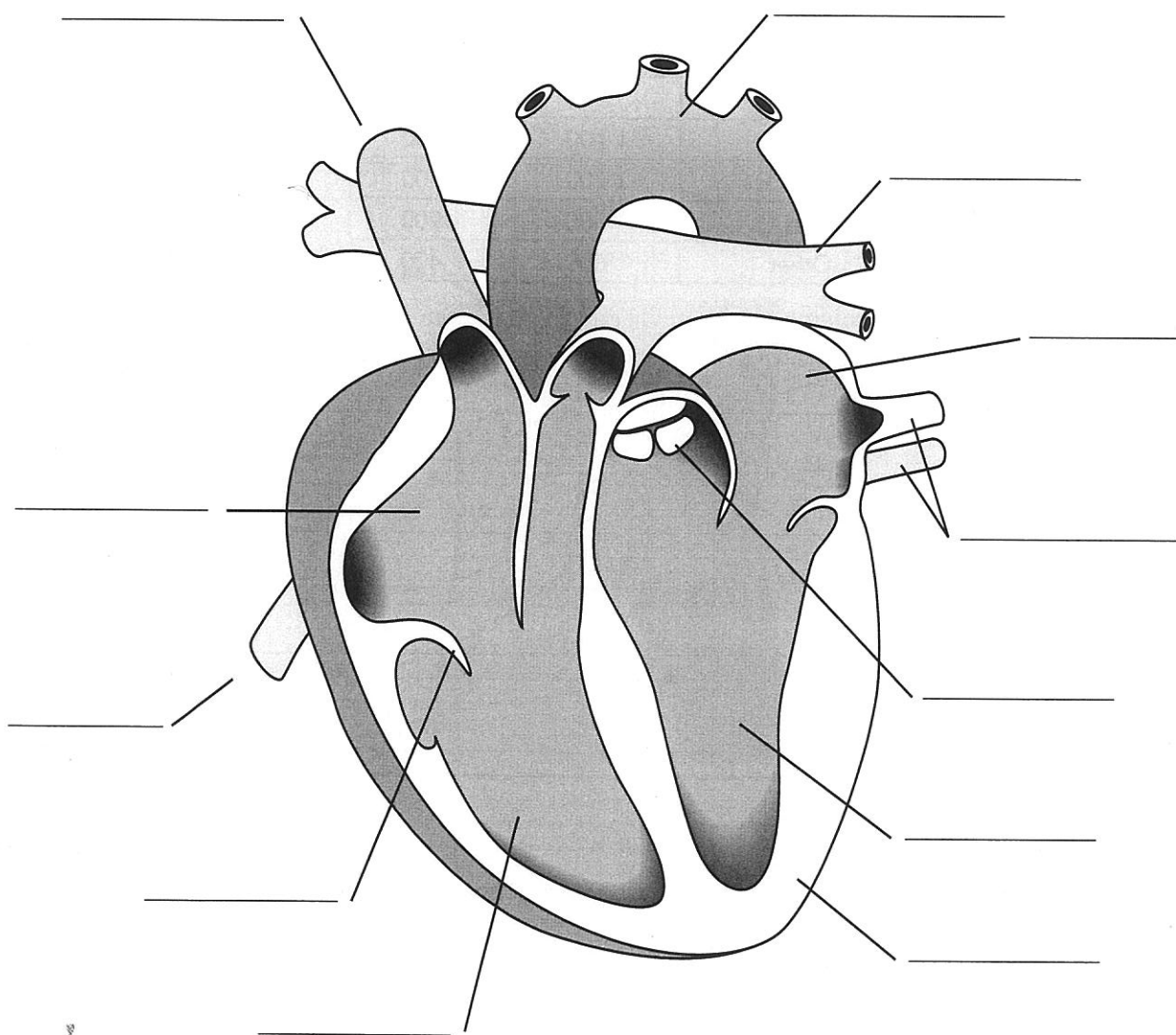


## The heart

Name: \_\_\_\_\_

**Skill: knowledge**

- 1 Label the parts of the heart.
- 2 Colour appropriate parts of the heart in red to show the path of oxygenated blood, and blue to show deoxygenated blood.
- 3 Show the direction of blood flow through the heart by drawing arrows.



- 4 Suggest why the left side of the heart has thicker muscle than the right.

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## Blood flow rates



Name: \_\_\_\_\_

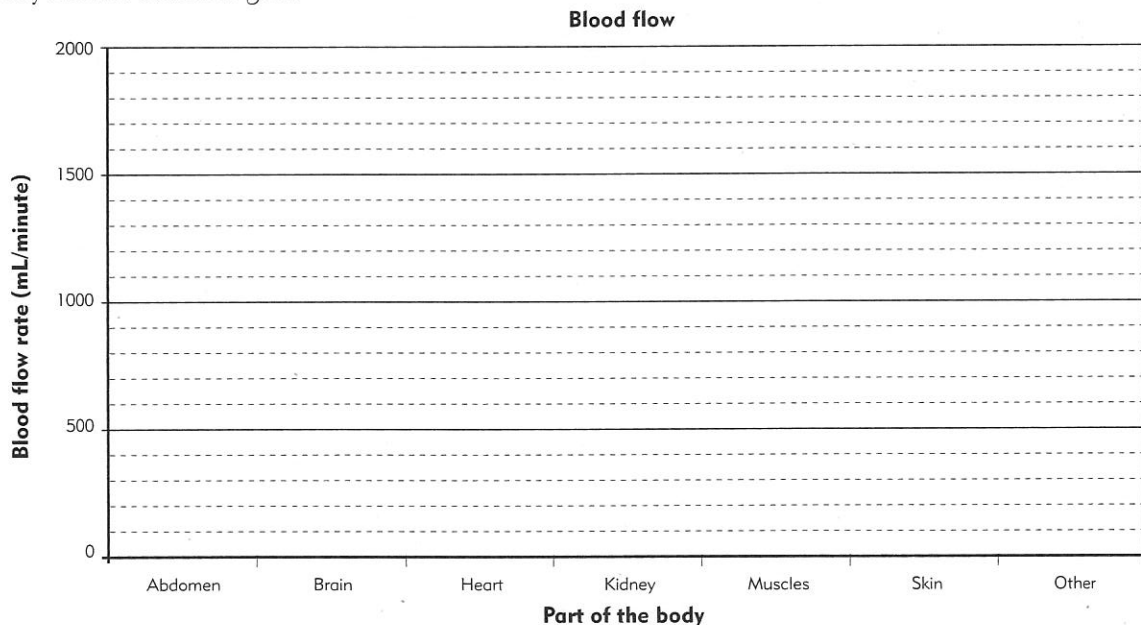
**Skills: interpreting, numeracy**

The rates of blood flow (in millilitres per minute) required by various parts of the body are listed below for an adult at rest and during strenuous exercise.

**Blood flow (mL/minute)**

| Body part | At rest | During strenuous exercise |
|-----------|---------|---------------------------|
| Abdomen   | 1 400   | 600                       |
| Brain     | 750     | 750                       |
| Heart     | 250     | 750                       |
| Kidney    | 1 100   | 600                       |
| Muscles   | 1 200   | 12 500                    |
| Skin      | 500     | 1 900                     |
| Other     | 600     | 400                       |

- 1 Show this information as column graphs on the grid below. Use an arrow and number to indicate a value beyond the available grid.



- 2 Which part of the body has the same blood flow whether the body is at rest or during strenuous exercise?  
\_\_\_\_\_
- 3 Which part of the body has the greatest change in blood flow when the body begins strenuous exercise?  
\_\_\_\_\_
- 4 What is the total volume of blood per minute pumped by the heart when the body is:
- a at rest? \_\_\_\_\_
- b strenuously exercising? \_\_\_\_\_